

Henry Nickless

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Education

University of Colorado, Boulder

Expected May 2028

Bachelor of Science in Aerospace Engineering Minor in Astrophysical and planetary sciences Cumulative GPA: 3.64/4.0

Projects

Taipan Liquid Rocket Engine (CU SRL)

August 2025 – Present

- Verified performance calculations for a 650lbf liquid bipropellant rocket engine, ensuring consistency between analytical models and expected thrust output
- Contributed to the redesign of an RP-1/LOX injector plate, performing iterative flow and combustion calculations while maintaining fixed engine geometry and a 700 psi chamber pressure constraint
- Applied NASA Chemical Equilibrium Analysis in Python to conduct parametric trade studies on mixture ratio, chamber pressure, and expansion ratio to evaluate performance sensitivity
- Developed Python-based optimization workflows leveraging AI-assisted methods to improve engine performance during propellant transition from ethanol/N₂O to RP-1/LOX
- Assisted in migrating test-stand software from Rust to C++ and authored embedded-systems scripts to ensure reliable hardware control during hot-fire testing

Regeneratively Cooled Liquid Rocket Engine (CU SRL)

January 2026 – Present

- Designed a 3D-printable unlike doublet injector plate with integrated igniter (ASI) ports and pressure sensors, optimizing structural integrity against combustion chamber constraints
- Applied topological optimization techniques to minimize component mass and maximize additive manufacturability
- Designed oxidizer-side stacked distribution rings and manifolds alongside gothic arches channels to maximize manufacturability
- Developed Python models to evaluate thrust, chamber pressure, and performance constraints.
- Performed preliminary heat transfer analysis for regenerative and film cooling strategies, estimating thermal loads and cooling effectiveness

Newtonian telescope for Astrophotography (@astro.htn)

December 2024 – Present

- Designed and 3D modeled 7+ telescope components to improve thermal stability and imaging performance without altering focal geometry
- Engineered temperature-resistant 3D-printed components with heat-set inserts for under \$30 total cost
- Designed and implemented an electronic dew-prevention cooling system for primary and secondary mirrors
- Designed CNC-milled focus adapters supporting camera systems exceeding 5 lbs

Glider Design Challenge

January 2026 – April 2026

- Collaborated with a six-member team to simulate and optimize glider performance using MATLAB
- Engineered an elevator deflection mechanism, achieving a 20° trim range while maintaining a sub-1-gram mass and minimal drag profile
- Validated design through flight testing, successfully transporting a 160g payload over a 100-meter distance

Skills

Fusion 360, SolidWorks, C++, MATLAB, Python, 3D printing, Excel, R, AI literacy

Work Experience

Merchandising sales associate and garden center team lead, LOWE's - Castle Rock, CO, May 2024 – August 2026

- Managed a team of 10+ people to complete time-dependent tasks while building clientele
- Helped to outline projects for customers and make sure they had everything to complete their project.

General Merchandising, Target – Castle Rock, CO, May 2023 – August 2023

- Managed multiple customer interactions and ensured a professional store appearance while adapting to change

Service Technician, Discount Tire – Centennial, CO, June 2022 – August 2022

- Completed time-oriented tasks while always maintaining store safety policies
- Interacted with customers to provide counsel according to their needs